

Zirong Chen

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EDUCATION

Vanderbilt University , Nashville, TN	Sept 2021 - May 2026
• <i>Doctor of Philosophy (Ph.D.)</i> in Computer Science; Advised by Dr. Meiyi Ma • <i>Research Interest</i>: Artificial Intelligence, Natural Language Processing, Large Language Models • <i>Dissertation Thesis</i>: Towards Trustworthy NLP-powered Intelligent Systems for Public Safety	
Georgetown University (M.S. in Computer Science), Washington, DC	Sept 2019 - May 2021
Jinling Institute of Technology (B.E. in Computer Science), Nanjing, China	Sept 2015 - May 2019

SELECTED AWARDS AND HONORS

IJCAI & AAAI Student Scholarship Awardee	2024, 2025
OpenAI Research Access Program Awardee (\$10,000)	Nov 2024
Vanderbilt Student Travel Grant Awardee (x3)	2023; 2024; 2025

SELECTED PUBLICATIONS

SenSys-2026 (submitted) **Zirong Chen**, Stephen Martini, Meiyi Ma. *Real-World Design and Deployment of an Embedded GenAI-powered 9-1-1 Calltaking Training System: Experiences and Lessons Learned*.

US Patent (provisional) **Zirong Chen**, Meiyi Ma. *A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Model Integration*. Filed in July, 2025.

AAAI-2026 David Hanchen Wang, Siwoo Bae, **Zirong Chen**, Meiyi Ma. *Learning with Preserving for Continual Multitask Learning*. Accepted at AAAI-2026 (17%) and **Selected Oral Spotlight**, Singapore, Singapore.

AAMAS-2025 Ziyan An, Xia Wang, Hendrick Baier, **Zirong Chen**, Abhishek Dubey, Taylor T Johnson, Jonathan Sprinkle, Ayan Mukhopadhyay, Meiyi Ma. *Combining LLMs with a Logic-Based Framework to Explain MCTS*. Published (39.9%) at AAMAS-2025, Detroit, MI.

IJCAI-2025 **Zirong Chen**, Ziyan An, Jennifer Reynolds, Kristin Mullen, Stephen Martini, Meiyi Ma. *LogiDebrief: A Signal-Temporal Logic Based Automated Debriefing Approach with Large Language Models Integration*. Accepted at IJCAI-2025 (13%), Montreal, Canada.

AAAI-2025 **Zirong Chen**, Elizabeth Chason, Noah Mladenovski, Erin Wilson, Kristin Mullen, Stephen Martini, Meiyi Ma. *Sim911: Towards Effective and Equitable 9-1-1 Dispatcher Training with an LLM-Enabled Simulation*. Published (18.9%) and **Selected Oral Spotlight Presentation (4.9%)** at AAAI-2025, Philadelphia, PA.

AAAI-2024 **Zirong Chen**, Xutong Sun, Yuanhe Li, Meiyi Ma. *Auto311: A Confidence-Guided Automated System for Non-emergency Calls*. Published (20.1%) and **Selected Oral Spotlight Presentation (6%)** at AAAI-2024, Vancouver, Canada.

Pervasive and Mobile Computing 2023 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec with shield: A secure intelligent assistant for requirement formalization*. Invited journal extension at Elsevier's Pervasive and Mobile Computing (2023)

IEEE-SmartComp-2022 **Zirong Chen**, Isaac Li, Haoxiang Zhang, Sarah Preum, John A Stankovic, Meiyi Ma. *CitySpec: An intelligent assistant system for requirement specification in smart cities*. Published and **Best Paper Finalist** at IEEE SmartComp 2022, Helsinki, Finland.

RESEARCH PROJECTS

Angie: AI-Driven Training System for 911 Call-takers in Emergency Responses	Dec 2023 - Present
<i>Integrating AI with real-world applications for a more efficient and effective emergency response training solution</i>	
• Led AI research project with Metro Nashville Emergency Communications, IT Services, and Government. Secured research grants from <i>OpenAI</i>, <i>Google</i>, <i>NSF(ReDDDoT)</i>, and <i>Vanderbilt</i>. • Served as major contributor and AI tech vendor, developing pioneering emergency response solution with	

sub-systems: *LogiDebrief*, *Sim911*, and *Auto311*. Deployed across 400+ workstations with 200+ active users, serving Metro Nashville for 14+ months.

LogiDebrief: An STL-based Automated 911 Call Debriefing System with LLMs Nov 2024 - Present
Disentangle overly complicated prompts into logic-enforced specifications with modularized use of LLMs

- Developed signal temporal logic (STL) framework integrated with LLMs to rigorously analyze and debrief historical 911/311 calls, maintaining accuracy across extended context windows. Through extensive real-world testing, LogiDebrief achieves over 95% accuracy without performance degradation on long contexts.
- Successfully deployed at local emergency response center for both daily operations and call-taker training programs. The system has automated analysis of over 1,701 emergency calls, resulting in 311+ hours of operational time savings.

Sim911: An LLMs-based Simulation System for Call-taker Training Mar 2024 - Present
An effective and reliable workflow for LLMs role-play simulation under emergency responses

- Developed context-aware generation and validation framework using CoT, RAG, and few-shot prompting with LLM agents to ensure realistic and inclusive simulations (addressing underserved groups). Framework incorporates automatic correction based on human feedback for continuous improvement.
- In real-world training programs, 90%+ participants rated Sim911 as comparable or superior to human-led simulations. Framework outperforms baseline by 62% in authenticity benchmarks.
- Deployed in local emergency response center's call-taker training programs. Generated 600+ simulated calls with 40+ hours of active engagement, saving 80+ hours of instructor time.

Auto311: A Confidence-Guided Automated System for Non-emergency Calls Jan 2023 - Present
Integrating AI with real-world applications for an automated call handling system

- Collaborated with Metro Nashville Department of Emergency Communications, Metro Nashville Information Technology Services, and Metro Nashville Government.
- Designed and developed an automatic call handling system for non-emergency calls based on confidence measurement with deep neural networks. Outperformed naïve LLMs calls with 94.49% acc under emulation. Implemented baseline models like Transformers.

INTERNSHIP EXPERIENCES

Research Fellow, Epiq AI Labs (formerly LAER AI) – New York, NY May 2025 – Aug 2025

- Enhanced AI eDiscovery AssistantTM with **DeepResearch** capabilities through agentic workflow design, optimized for eDiscovery, document review, and complex legal scenarios.
- Created **BREEW** (Bayesian Reasoning via Entity-Evidence Weaving) benchmark that leverages concept decomposition and entity-level evidence surfacing to ensure logical coherence and rigorous concept validation.
- Built **LINT** (Learning via Interactive Tutoring) testing environment featuring simulated user agents with restorable states. The system evaluates agent performance through controlled conversational interactions using probabilistic finite state machines and configurable personas.

Data Scientist(NDA) , National Basketball Association (NBA) – New York, NY May - Aug 2020, 2021

- Analyzed user feedback using neural networks (PyTorch, Keras) and NLP tools (NLTK, spaCy) to develop data-driven fan engagement strategies.
- Enhanced marketing campaign effectiveness by applying statistical ML models (scikit-learn) to optimize user experience and advertisement targeting.
- Developed real-time data analysis pipelines with sequential neural networks (RNNs, Time2Vec) to predict streaming bandwidth requirements, optimizing infrastructure costs and user experience.
- Architected scalable and reusable neural network frameworks for cross-project deployment, improving development efficiency and model adaptability.

ML Engineer(NDA) , Sugon, Chinese Academy of Sciences – Nanjing, China Dec 2018 - May 2019

- Developed projection algorithms based on vectorized path trajectories and distance metrics (Fréchet, Manhattan) to improve GPS accuracy in dense urban areas.
- Created dynamic routing algorithms using hidden Markov models to optimize urban planning and navigation tasks.

- Contributed to a high-impact, data-driven project involving predictive modeling and real-time decision-making for urban management applications.

SELECTED TALKS AND SERVICES

Invited Talk on “LLM-based Role Playing for Emergency Response Training” by Dr. Ehsan Hoque’s ROC-HCI Lab at Unverisity of Rochester	October 2025
Invited Educational AI Demo at LIVE Learning Innovation Incubator	Mar 2024, 2025
Guest lectures at Vanderbilt (Cyber Physical AI): “Large Language Models”	2022; 2023; 2024; 2025
Guest lectures at Vanderbilt (Artificial Intelligence): “Natural Language Processing”	2022, 2023, 2024
Conference/Journal Program Committee (e.g., ICLR, AAI, ACM Trans, etc.)	